Fiscal Year:	FY 2008	**Task Last Updated:**	FY 10/31/2008
PI Name:	Vander Wal, Randall L. Ph.D.		
Project Title:	Quantification of Fire Signatures for Practical Spacecraft Materials		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44135-3191	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	01/01/2003	**End Date:**	09/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Project end date extended until 9/30/2008, per T. Rusnak/GRC (3/08--end date amended 6/08 after realized misinterpretation) NOTE: Project end date extended until 6/30/2008, per T. Rusnak/GRC (3/08) NOTE: Funding received late; project continues until 9/30/2007, per N. Hall/GRC (5/07) NOTE: Added to Task Book 10/06 per GRC info		
Key Personnel Changes/Previous PI:	Different personnel were involved in the course of the project and each is listed as a Co-I for their contributions.		
COI Name (Institution):	Ruff, Gary (NASA GRC) Novak, Jane Hitomi (NSCER) Pushkarev, Vladimir (NCSER)		
Grant/Contract No.:	In-House		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

Fire protection on the Space Shuttle and International Space Station has been developed primarily through the modification of standard practices in terrestrial fire safety applications and experience gained throughout the history of the U.S. manned space program. The primary feature of a spacecraft fire protection strategy is material control and elimination of potential ignition sources. Underlying this strategy is that any ignition event that does start a fire will remain "small" and not spread until the fire is detected and the crew has time to respond. The fact that the ISS has only portable fire extinguishers is testimony to the role of the crew in the fire protection system. In the 20 years of Shuttle operation, there have been five potential fire-causing incidents of component overheating or electrical short-circuiting. In all of these incidents, the crew recognized and corrected the problem before the smoke detector actuated. While these incidents can be considered a success for the fire protection system, one has to wonder how long the event would have continued before the smoke detectors reached the alarm threshold. The current era of larger spacecraft with fewer crew members increases the likelihood that a pre-fire incident will occur in an unoccupied module.
Obviously, if the crew can detect the fire before the smoke detectors, fire precursors are present in the atmosphere that an appropriately designed and placed sensor could detect. The trick is knowing what compound or compounds to look for and in what concentrations. Smoldering and pyrolysis, common pre-fire events, generate substantial yields of volatiles (partial decomposition products) that could be used as pre-fire signatures. There are currently at least three research projects aimed at developing techniques for improved fire detection, all based on the measurement of gaseous products of combustion such as CO, NOx, HCN, HCl, and total hydrocarbons. These include several SBIR projects developing sensitive laser absorption techniques as well as the development of MEMS gas sensors being conducted at the NASA Glenn Research Center. While these sensors are promising for use in spacecraft because of their small size and low power requirements, a more basic problem exists – to make use of these sensors to detect a fire, one must know what chemical species and concentrations to look for.

Studies conducted in 1-g environments have measured the yield and identity of partial decomposition products as function of the rate of heating and oxygen concentration. Though insightful, buoyancy convolves the rate of heating with oxidizer entrainment and combustion product removal. Thus variability in partial decomposition products (identity and yield) and occurrence of secondary reactions of evolved volatiles can be obscured. The non-buoyant or weakly convective flows characteristic of a microgravity environment will alter the rates of oxidizer entrainment and combustion product removal. The tenet of this proposal is that the specific chemical reactions (and their products) can differ substantially in such environments. With smoldering or pyrolysis events as characteristic pre-fire and "fire" events, variability in the evolved partial decomposition products and their concentration has consequences for both fire detection and toxicological assessment.

The proposed study seeks to investigate the differences (identities and relative concentrations) of the volatiles produced by pyrolyzing and/or smoldering materials in a microgravity environment. Test materials will be representative of typical spacecraft materials and, where possible, will be tested in appropriate geometries. Wire insulation materials of Teflon, polyimide, silicone, and PVC will be evaluated using either cylindrical samples or actual wire insulation. Other materials such as polyurethane, polyimide, melamine, and silicone-based foams will be tested using cylindrical samples, in addition to fabric materials, such as Nomex. Electrical components, such as resistors, capacitors, circuit board will be evaluated.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Several design considerations for both detectors, fire suppression and materials (both structural and cosmetic) have been illuminated by these studies.
Specific recommendations follow:

Materials that evolve substantial particulate or CO under any conditions of pyrolysis of smoldering present either a direct health risk under such conditions or during extinguishment actions. Production of organic compounds such as carboxylic acids or inorganic acids such as HF or HCl will cause corrosion and quickly render environments non-habitable at very low (10’s ppm) concentrations. Given the low levels for toxicity associated with HF and HCl, coupled with relatively large amounts of such materials aboard a confined volume, advanced alternative wire insulation materials ought to be sought. So-called oily smokes or aerosols will pose daunting cleanup challenges. Other materials producing aerosols or particulate of unknown composition or with morphologies analogous to nanomaterials with known biological effects could pose long-term chronic toxicity. Advanced materials could mitigate these risks. For example, new clay-polymer or carbon-polymer composites not only have higher glass transition temperatures and depressed heat release rates but can even be rendered inflammable. Moreover our own measurements to be reported elsewhere have shown that release of toxic byproducts from smoldering or pyrolysis to be substantially mitigated by incorporation of these nanofiller materials.

Suppression activities ought to be thoroughly investigated. Altered temperatures and ambient environments will change pyrolysis and combustion modes leading to potentially drastic changes in byproduct identities and concentrations. For the materials studied here fire suppression could create a more hazardous environment than the original fire. Fortunately the effects associated with such transient actions associated with suppression can be adequately tested in low-gravity emulators such as drop towers and aircraft.

Side effects associated with particulate matter must be considered. Organic and elemental carbon particulate can clog filters and coat the interior of air-handling systems. Moreover high smoke concentrations can limit if not eliminate visibility and thereby inhibit or prevent evacuation.

Finally the health effects of these materials ought to be evaluated. The tar-like compounds could contain aromatic or highly substituted aromatics possessing –F, -OH, -COOH or other functional groups. Such chemical functionalities are known to be irritating to lung and epithelial tissue.

Task Progress:	Introduction In complimentary work Thermal Gravimetric Analysis – Fourier Transform Infrared (TGA-FTIR), a so-called hyphenated technique was used to identify the gases evolved during smoldering and flaming combustion of polymers approved for spacecraft use. Performed at a constant heating rate, gases evolved during the temperature ramp are detected by FTIR. Comparison against reference library spectra permits identification of the various functional groups of gaseous components and molecular identification of smaller species. In this work segment FTIR spectra were collected at several points along the TGA curve beginning with the initiation of sample weight loss and at subsequent points in the temperature ramp. Conclusions are drawn summarizing the changes which occur in the stages of decomposition of the resins. Conclusions Evolved fire signatures from the combustion of six polymers approved for spacecraft were determined through two techniques. The gases produced at different stages of the fire were identified by TGA/FTIR and the quantification of specific gases was performed after the smoldering and flaming of these materials in the vertical furnace following the chromatographic analysis of the gas sample. Among the great number of gaseous products identified by IR and quantified by GC, CO2 appears to be the predominant species corresponding to the complete thermal degradation in the atmosphere of air. However, this signature is produced in the middle and end of a fire event. We observed that clearly no single species serves as a consistent indicator for the detection of the initial fire. The most conservative sensor design should take in account the detection of more than one fire signature.
Bibliography Type:	Description: (Last Updated: 03/08/2019)
Articles in Peer-reviewed Journals	Vander Wal RL, Pushkarev V, Novak JH-F. "Fire signatures of spacecraft materials: Gases and particulates." Combustion and Flame. 2012 Feb;159(2):897–904. http://dx.doi.org/10.1016/j.combustflame.2011.08.012 , Feb-2012
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Papers from Meeting Proceedings	Vander Wal RL, Novak JH-F. "Fire Signatures of Spacecraft Materials: Gases, Condensables and Particulates." Central States Section,The Combustion Institute, 2008 Technical Spring Meeting, Tuscaloosa, AL April 20th – 21st, 2008. Central States Section,The Combustion Institute, 2008 Technical Spring Meeting, Tuscaloosa, AL April 20th – 21st, 2008. , Apr-2008